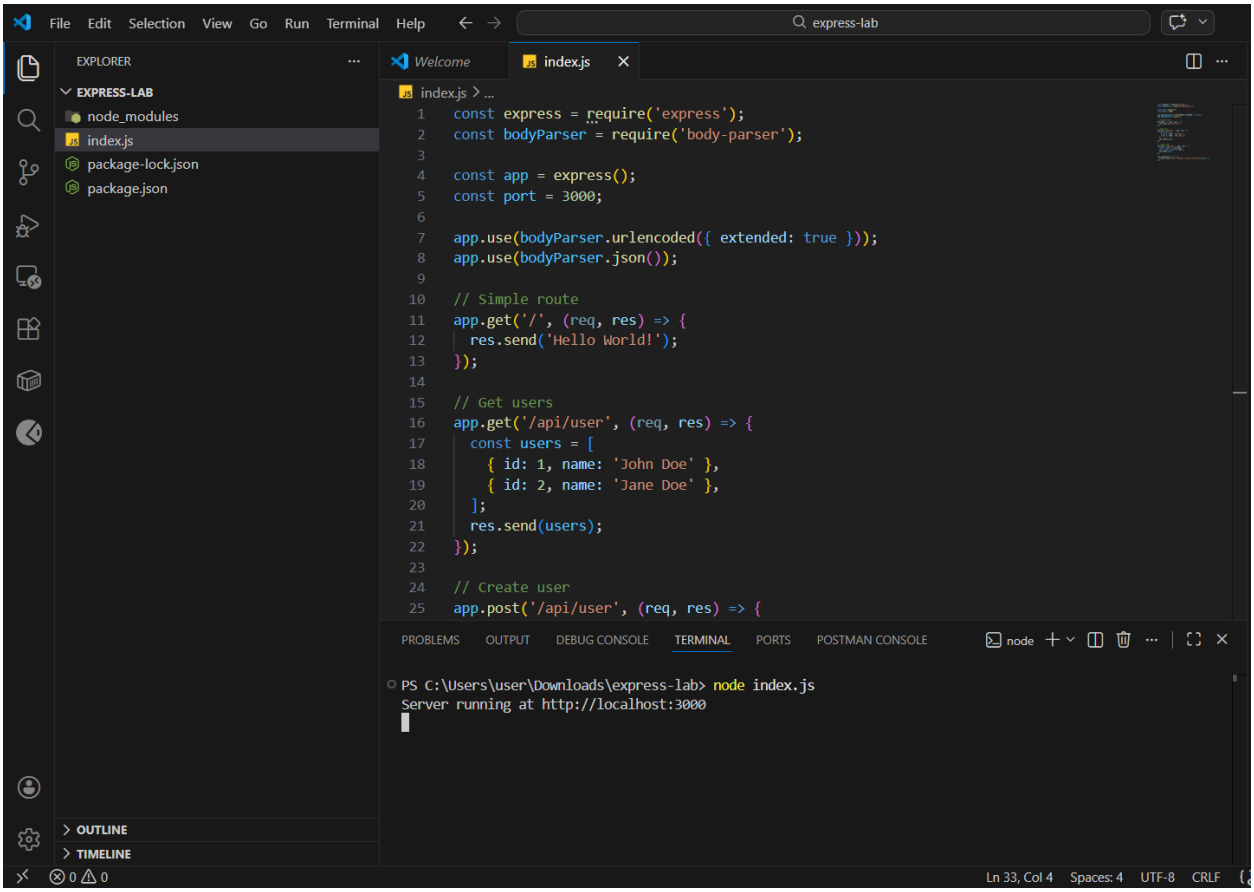
SE3040 – Application Frameworks
BSc (Hons) in Information Technology
Software Engineering Specialization
3rd Year
Faculty of Computing
SLIIT
2026 - Practical
Lab 04

Lab session 4 – Express.js REST API.

```
PS C:\Users\user\Downloads> mkdir express-lab


    Directory: C:\Users\user\Downloads


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/22/2026  10:36 AM                express-lab


PS C:\Users\user\Downloads> cd express-lab
PS C:\Users\user\Downloads\express-lab> npm init -y
Wrote to C:\Users\user\Downloads\express-lab\package.json:

{
  "name": "express-lab",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}



PS C:\Users\user\Downloads\express-lab> npm install express body-parser

added 65 packages, and audited 66 packages in 3s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```


File Edit Selection View Go Run Terminal Help
express-lab
EXPLORER
EXPRESS-LAB
node_modules
index.js
package-lock.json
package.json
Welcome
index.js
index.js > ...
const express = require('express');
const bodyParser = require('body-parser');
const app = express();
const port = 3000;
app.use(bodyParser.urlencoded({ extended: true }));
app.use(bodyParser.json());
// Simple route
app.get('/', (req, res) => {
res.send('Hello World!');
});
// Get users
app.get('/api/user', (req, res) => {
const users = [
{ id: 1, name: 'John Doe' },
{ id: 2, name: 'Jane Doe' },
];
res.send(users);
});
// Create user
app.post('/api/user', (req, res) => {
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
node
PS C:\Users\user\Downloads\express-lab> node index.js
Server running at http://localhost:3000
OUTLINE
TIMELINE
Ln 33, Col 4 Spaces: 4 UTF-8 CRLF

Hello World!

http://localhost:3000/

Save | Share

GET http://localhost:3000/ Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body | Cookies | Headers 7 | Test Results | 200 OK • 109 ms • 239 B

HTML | Preview | Visualize

```
Hello World!
```

http://localhost:3000/api/user

Save | Share

GET http://localhost:3000/api/user Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body | Cookies | Headers 7 | Test Results | 200 OK • 21 ms • 290 B

JSON | Preview | Visualize


```
[
    {
        "id": 1,
        "name": "John Doe"
    },
    {
        "id": 2,
        "name": "Jane Doe"
    }
]
```

http://localhost:3000/api/user

Save | Share

POST http://localhost:3000/api/user Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
{
  "name": "Sakith"
}
```

Body | Cookies | Headers 7 | Test Results | 200 OK • 89 ms • 259 B

JSON | Preview | Visualize

```
{
    "id": 3,
    "name": "Sakith"
}
```

EXPLORER

EXPRESS-LAB
- node_modules
- index.js
- package-lock.json
- package.json

OUTLINE
TIMELINE

Welcome | index.js

index.js > ...

```
const express = require('express');
const bodyParser = require('body-parser');

const app = express();
const port = 3000;

app.use(bodyParser.urlencoded({ extended: true }));
app.use(bodyParser.json());

let todos = [
  { id: 1, text: 'Buy groceries', done: false },
  { id: 2, text: 'Do laundry', done: true },
];

// GET all todos
app.get('/api/todos', (req, res) => {
  res.send(todos);
});

// GET single todo
app.get('/api/todos/:id', (req, res) => {
  const id = Number(req.params.id);
  const todo = todos.find(todo => todo.id === id);

  if (todo) {
    res.send(todo);
  } else {
    res.status(404).send('Todo not found');
  }
});

// POST new todo
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | POSTMAN CONSOLE | node

```
PS C:\Users\user\Downloads\express-lab> node index.js
Server running at http://localhost:3000
```

0 ⚠ 0 | Ln 9, Col 1 | Spaces: 4 | UTF-8 | CRLF

http://localhost:3000/api/todos

Save | Share

GET http://localhost:3000/api/todos Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

Body | Cookies | Headers 7 | Test Results | 200 OK • 27 ms • 322 B

{} JSON | Preview | Visualize


```
[
    {
        "id": 1,
        "text": "Buy groceries",
        "done": false
    },
    {
        "id": 2,
        "text": "Do laundry",
        "done": true
    }
]
```


http://localhost:3000/api/todos/1

Save | Share

GET http://localhost:3000/api/todos/1 Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

Body | Cookies | Headers 7 | Test Results | 200 OK • 7 ms • 279 B

{} JSON | Preview | Visualize


```
{
    "id": 1,
    "text": "Buy groceries",
    "done": false
}
```

http://localhost:3000/api/todos

Save | Share

POST http://localhost:3000/api/todos Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2   "text": "Study Express",
3   "done": false
4 }
```

Body | Cookies | Headers 7 | Test Results | 200 OK • 9 ms • 279 B

{} JSON | Preview | Visualize

```
1 {
2   "id": 3,
3   "text": "Study Express",
4   "done": false
5 }
```

http://localhost:3000/api/todos/1

Save | Share

PUT http://localhost:3000/api/todos/1 Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2   "text": "Updated task",
3   "done": true
4 }
```

Body | Cookies | Headers 7 | Test Results | 200 OK • 6 ms • 277 B

{} JSON | Preview | Visualize

```
1 {
2   "id": 1,
3   "text": "Updated task",
4   "done": true
5 }
```

http://localhost:3000/api/todos/1

Save | Share

DELETE http://localhost:3000/api/todos/1 Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1

Body Cookies Headers 7 Test Results 200 OK 9 ms 250 B

HTML Preview Visualize

1 Todo with id 1 deleted